



dd/mm/2025

Opportunity meeting

Contextual Bandit Reporting

Objective

1

Problem statement

2

Why it should be addressed?

3

Competition

4

Proposed solution

5

Open discussion

6

Decision

Agenda

Gain buy-in for redesigning CB reporting to
make it factual & understandable

Problem statement

PROBLEM STATEMENT

User Persona

- **Customer Type:** All enterprise customers running advanced experiments
- **Product offer:** Web Experimentation & Feature Experimentation
- **Industry:** Cross-industry
- **User segment:** Broad range, mainly analysts & PMs
- **Specific criteria:** Users leveraging Contextual Bandits or Multi-Armed Bandits (MAB) for live traffic allocation
- **Customers names or reporters:** Fred
- **Reach (#people impacted):** 999 (~X%) of XPs, 27 accounts

PROBLEM STATEMENT

Problem description

As any data person,
when I want to understand why the bandit reallocated traffic or preferred a given variant,
I get the following problem: the report looks identical to a normal A/B test and shows misleading “confidence” metrics that don’t apply to such adaptive algorithms.

Today, we can manually explain results via support or CSM calls, but this is a huge CX pitfall that is not scalable and erodes trust in our automated bandits’ reporting and even overall statistical prowess.

Why should we address the problem?

Opportunity type & justification

Opportunity types	Justification examples
Improvement to existing workflow	• Frequent confusion about reliability of CB/MAB results • Support time spent explaining CB results cannot scale • JC confirms data reporting is inaccuracies. "Yeah I think it's even worse than that, the current numbers we report are just wrong 😞"

WHY

Which OKR is connected?

Explain how this opportunity is connected to your OKRs.

Linked OKR	Why?
Indicate which OKR is connected to this opportunity. Add one row for each involved OKR.	Explain why it is connected. Detail your answer with bullet points.
{ REPLACE }	{ REPLACE }

WHY

Expected impact

Use the previous slide as a reference to fill out this one

OKR contribution

{ **X %** - Estimate the percentage contribution of this initiative to your OKR achievement. }

User benefits impact

{ **X / 5** - Rate the expected impact on a scale from 1 to 5 }

Business impact

{ **X / 5** - Rate the expected impact on a scale from 1 to 5 }

Competition

Comp Benchmark

Comp	Solution Summary	Method	CMAB?	★
SS	Reporting focuses on predicted performance & confidence bounds. Supports binary, numeric, and continuous metrics	LinUCB (CMAB)	✓	5
GB	Only has classic MAB. Shows a leaderboard ranking variations by observed reward, win rate, and confidence bounds; updates dynamically to reflect cumulative results.	Thompson Sampling (MAB only)	✗	5
LD	Displays cumulative conversion and reward results alongside traffic reallocation trends; combines learning visualization and rollout-style reporting	Bayesian MAB / Thompson Sampling	✗	5
OPT	Frames results as improvement over original or holdout. Avoids significance. Optimizely developed tree-based models for binary classification.	Proprietary Bayesian MAB + CMAB	✓	5
Eppo	Runs holdout experiments comparing CMAB vs control; separates “Learning Mode” and “Evaluation Mode” for transparent lift measurement	Linear-model CMAB + Holdout	✓	4
ABT	Not enough info	Undisclosed	✗	?
VWO	Allocate traffic and minimize regret; dashboard highlights conversion and traffic trends instead of significance. Documentation is too technical and all over the place.	Thompson Sampling	✗	2

Compared to our competitors, we are lagging

Optimizely

Their RP displays real-time traffic allocation and performance metrics for each variation, ranked by their estimated reward. Possibly employs a holdout to measure lift compared to **equal split**, avoids statistical significance.

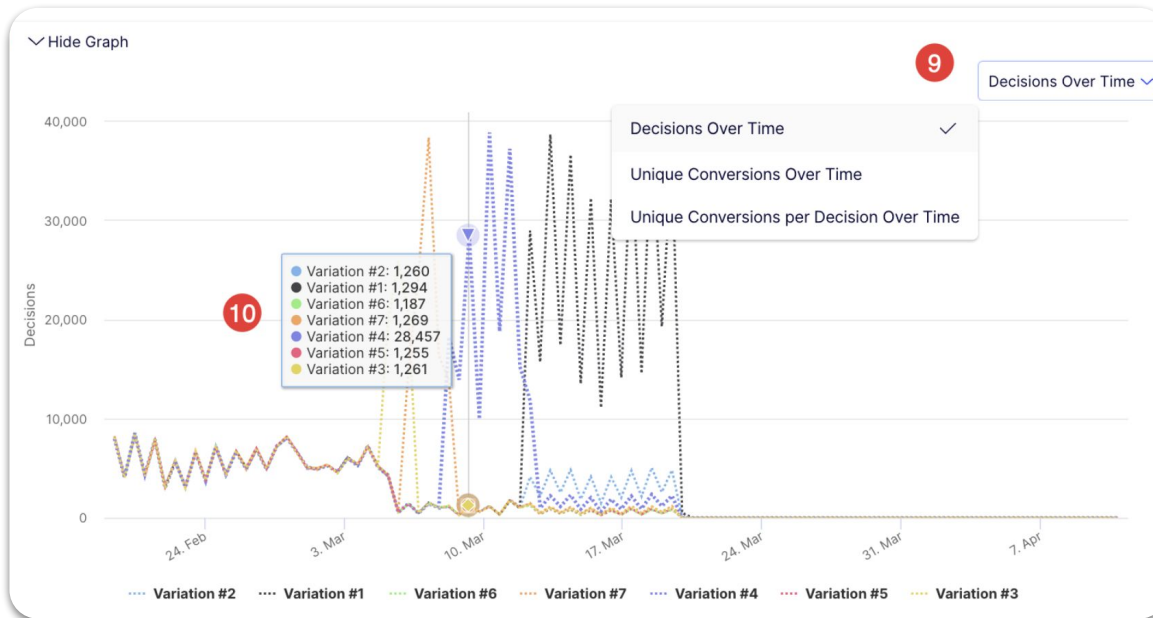
Other than tables there is just this time chart showing allocation changes over time.

<https://support.optimizely.com/hc/en-us/articles/35658523190541-Contextual-Bandit-Results-page>

<https://support.optimizely.com/hc/en-us/articles/4410289035405-Maximize-lift-with-multi-armed-bandit-optimizations>

Top User Attributes ?

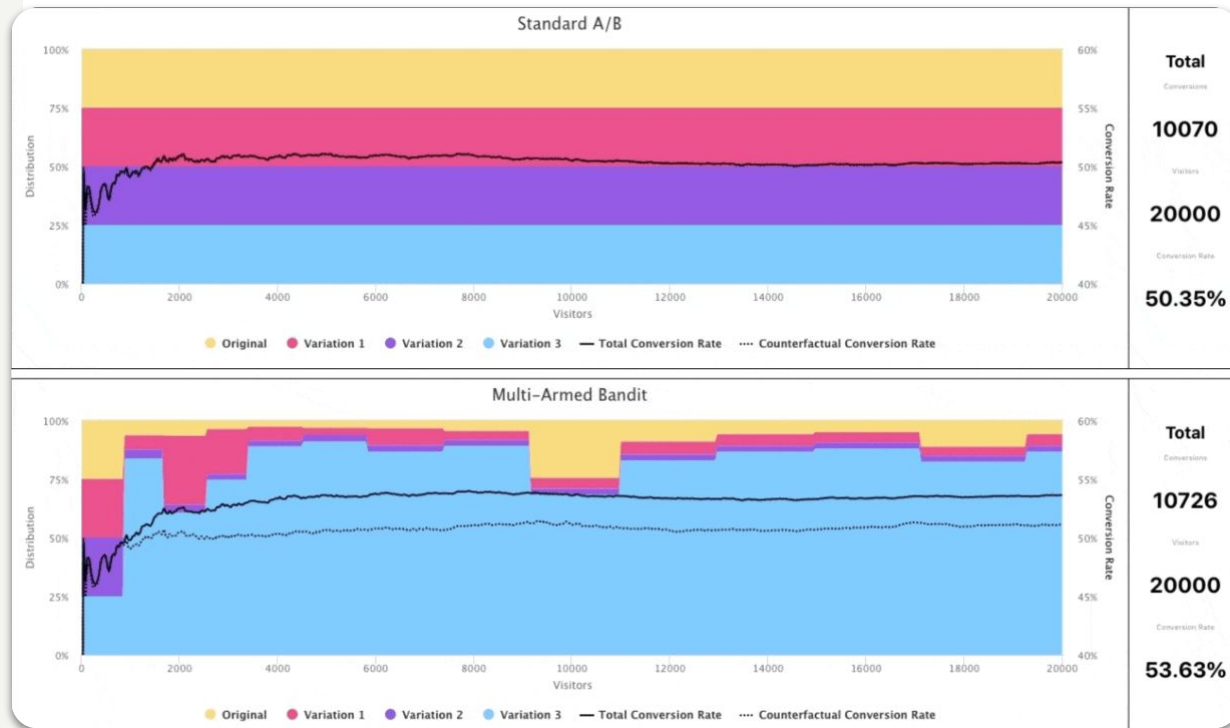
#	Name	Impact
1	Device type	27%
2	Product type	8%
3	User role	8%
4	Language	6%



Optimizely

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Notably helpful is this chart showing allocation changes over time.



<https://support.optimizely.com/hc/en-us/articles/35658523190541-Contextual-Bandit-Results-page>

<https://support.optimizely.com/hc/en-us/articles/4410289035405-Maximize-lift-with-multi-armed-bandit-optimizations>

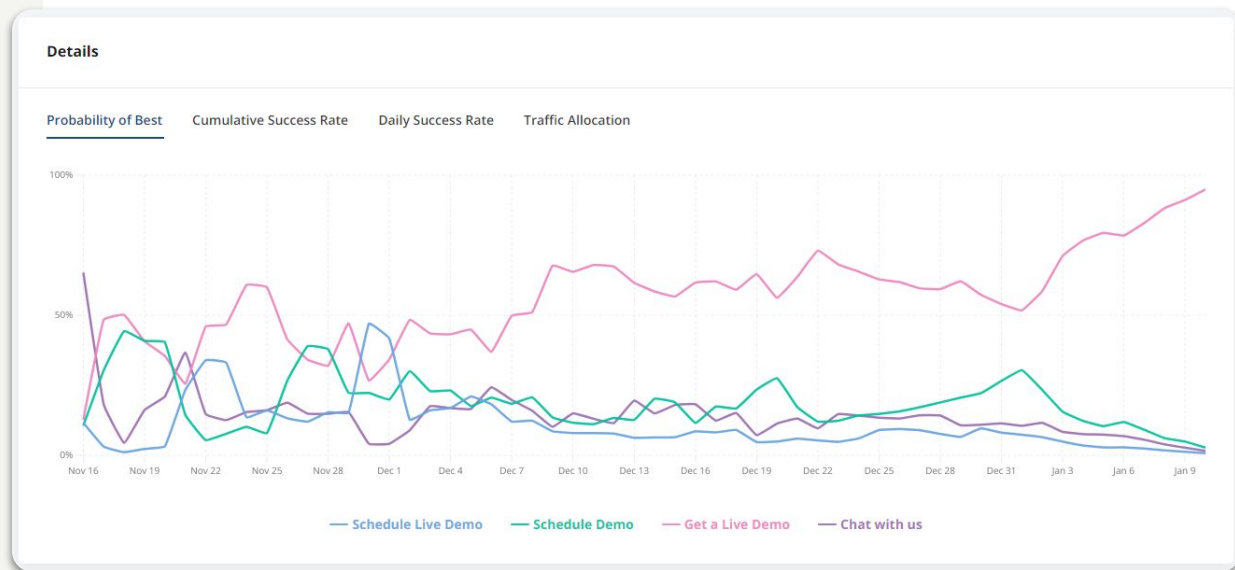
Statsig

Autotune AI selects the best variant using [LinUCB](#) logic with **built-in confidence intervals**.

The results interface focuses on predicted outcomes and exploration vs exploitation balance, not statistical significance.

Technically this is the closest method to MLAI any competitor employs for their CMAB.

<https://docs.statsig.com/autotune/contextual/introduction>



Probability of Best

Cumulative Success Rate

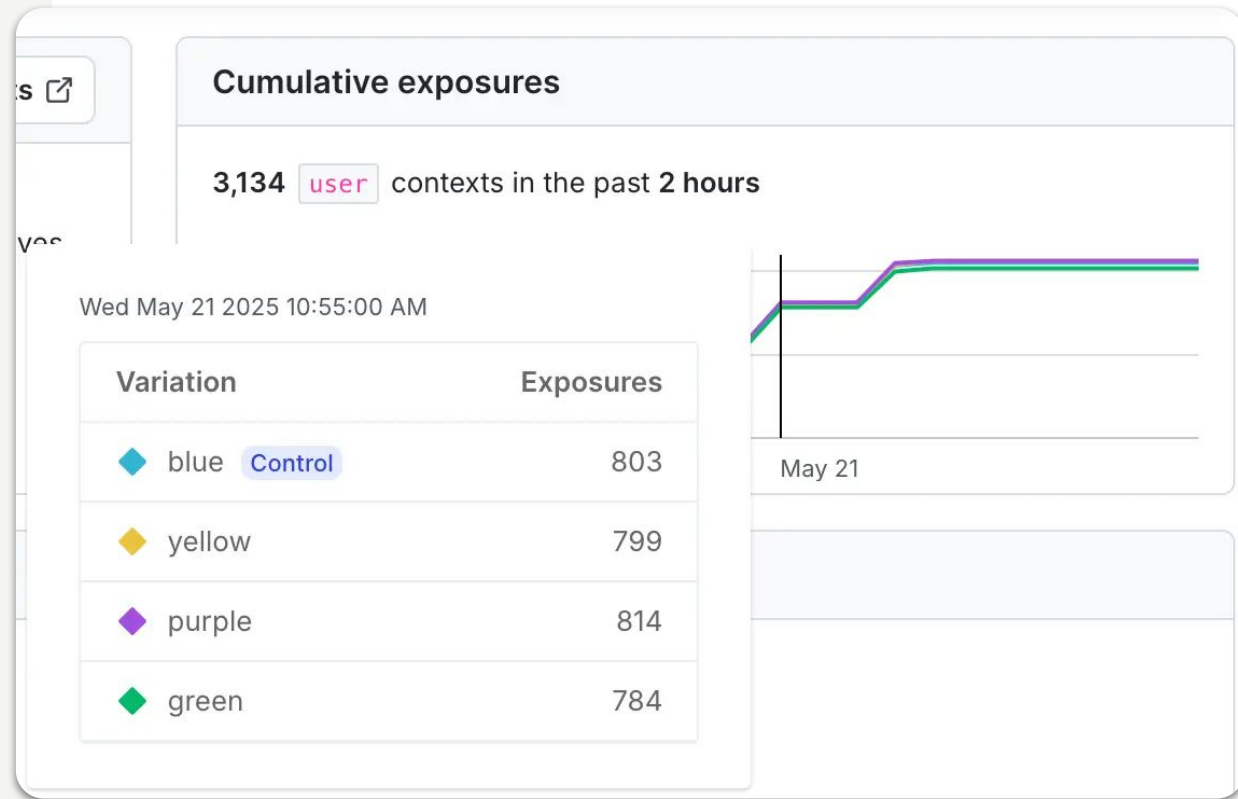
Daily Success Rate

Traffic Allocation

LaunchDarkly

Results view combines **cumulative performance** reporting with real-time allocation tracking.

It shows how traffic is distributed among variations and summarizes **cumulative** conversion rates and rewards across the experiment.



<https://launchdarkly.com/docs/ho-me/multi-armed-bandits/results>

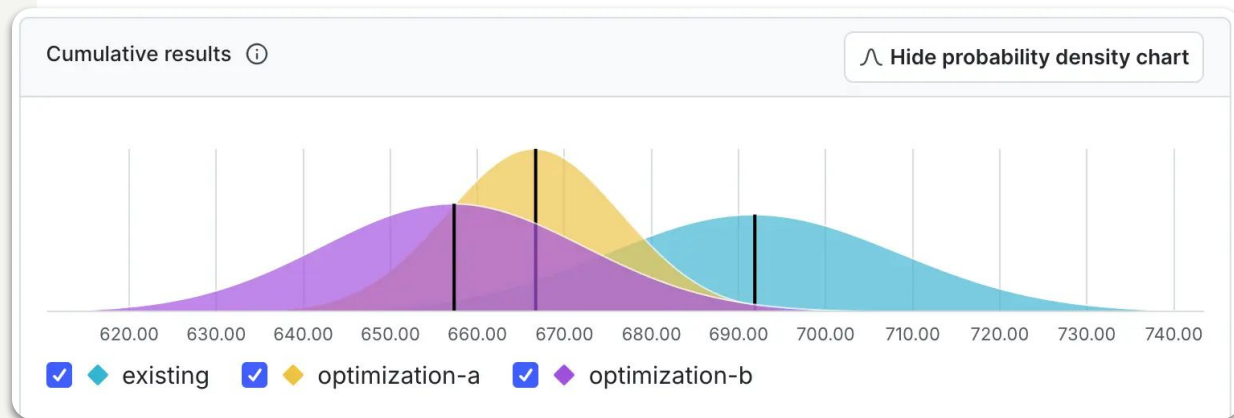
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Cumulative results ⓘ					⌵ High probability density chart
Variation	Probability to be best	Conversion rate	Conversions	Exposures	
◆ blue	2.50%	50.81% [47.79, 53.82]%	535	1,053	
◆ purple	9.52%	52.05% [49.07, 55.01]%	553	1,062	
◆ green Leading variation	87.98%	54.77% [51.74, 57.78]%	559	1,019	

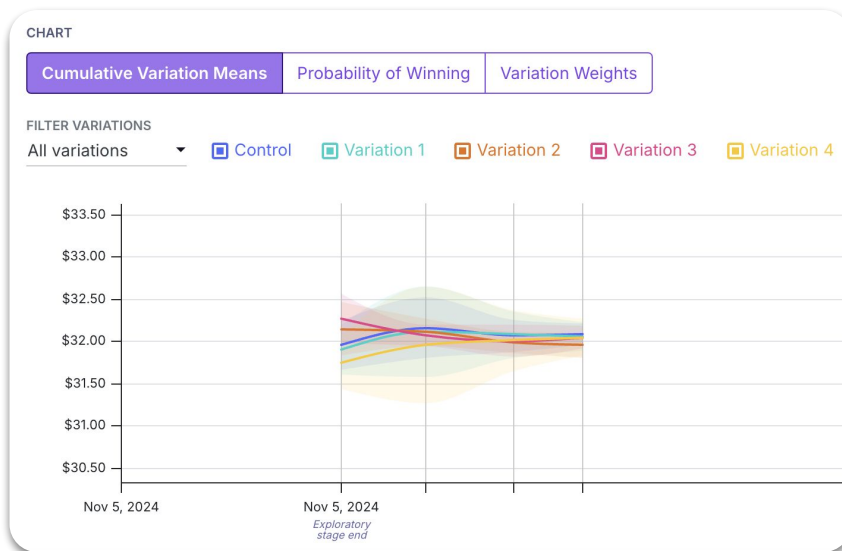
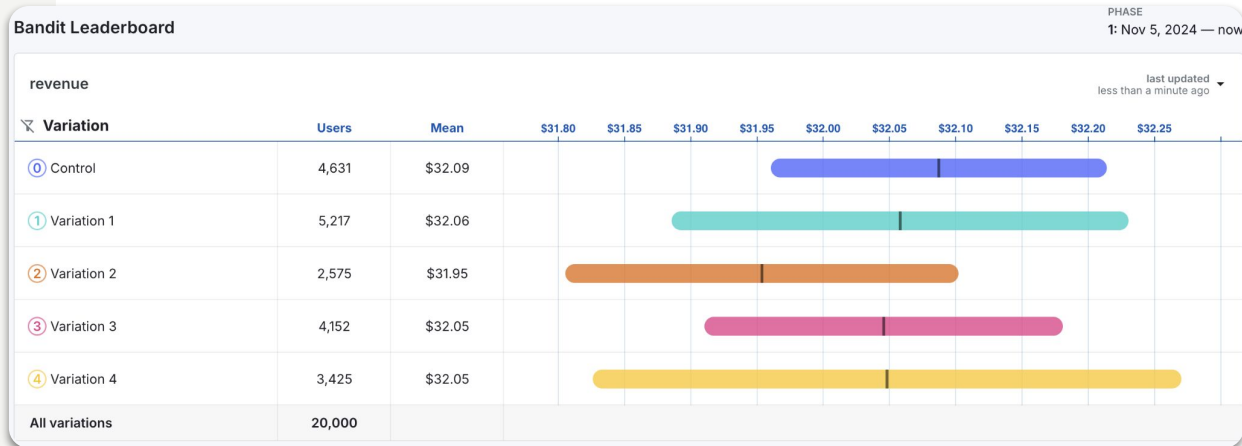


Growthbook

Their RP shows a **leaderboard** ranking all variants by performance, including observed win rates, cumulative rewards, and confidence bounds based on Thompson Sampling.

The interface updates **dynamically** as more data comes in and highlights the leading variant visually.

<https://docs.growthbook.io/bandit/s/results#leaderboard>

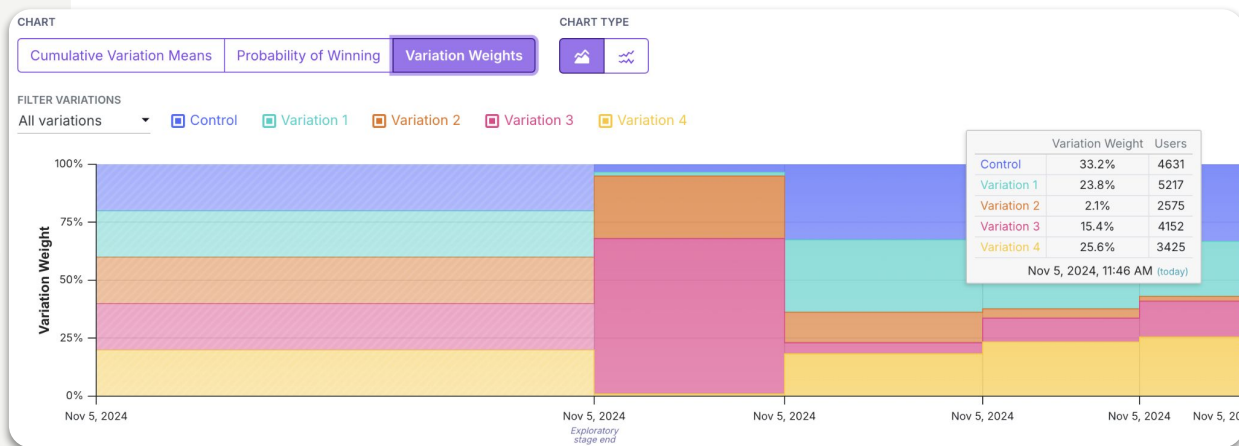
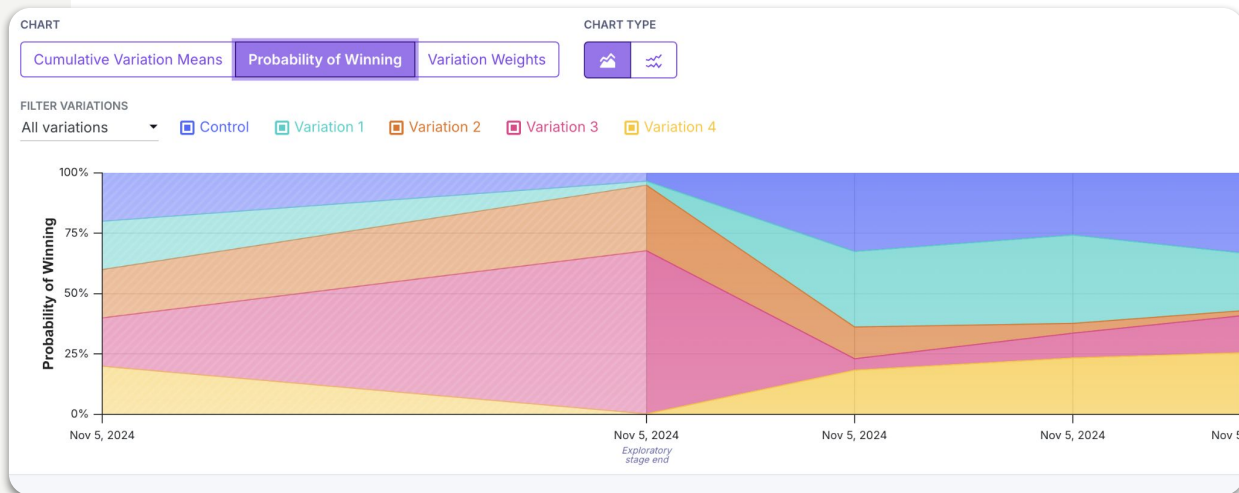


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Eppo

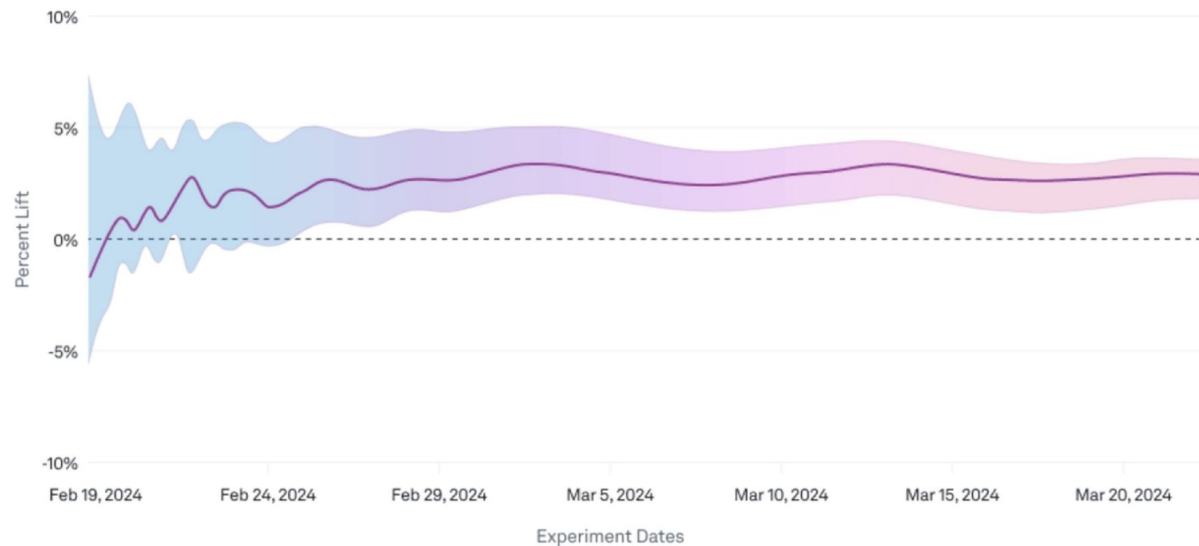
Eppo measures the effectiveness of CBs by running them side-by-side with a control group (**holdout experiment**).

The holdout group doesn't use the bandit, allowing Eppo to measure **true lift** i.e how much better the bandit performs compared to traditional allocation.

<https://docs.geteppo.com/contextual-bandits/#measuring-impact-using-experiments>

Banner Click-Through Rate

2.59%



PERCENT LIFT + CONFIDENCE INTERVAL



Ad Display Bandit

.....



Age-Based Display

AB Tasty

Very little info available.

Dynamic allocation is the only term used and has a basic description with no visuals I could find.

<https://docs.abtasty.com/web-experimentation-and-personalization/traffic-allocation/dynamic-allocation>

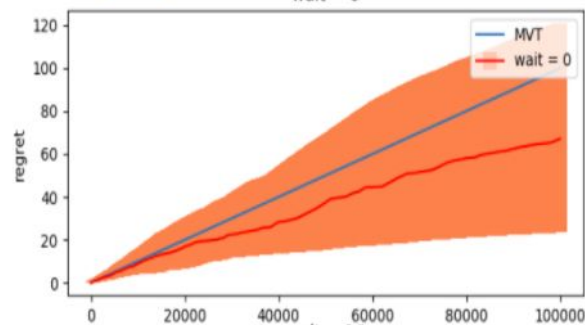
VWO

VWO's Multi-Armed Bandit automatically increases traffic to better-performing variations using the **Thompson Sampling** algorithm.

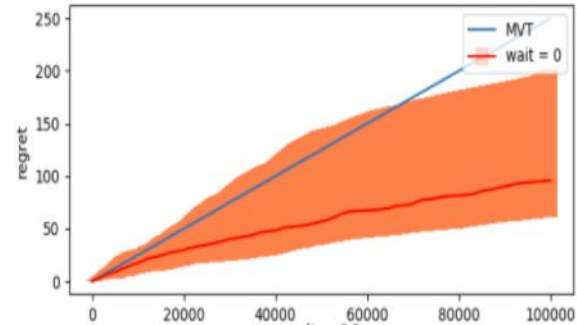
It aims to **minimize regret** i.e the loss from sending visitors to underperforming variants.

<https://help.vwo.com/hc/en-us/articles/900003845666-Understanding-the-Working-of-Multi-Armed-Bandit-in-VWO>

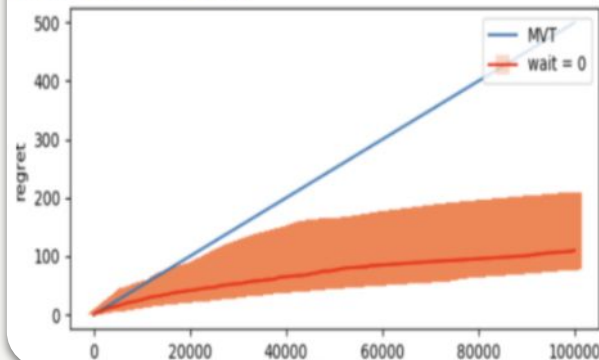
10% CR, 2% Uplift



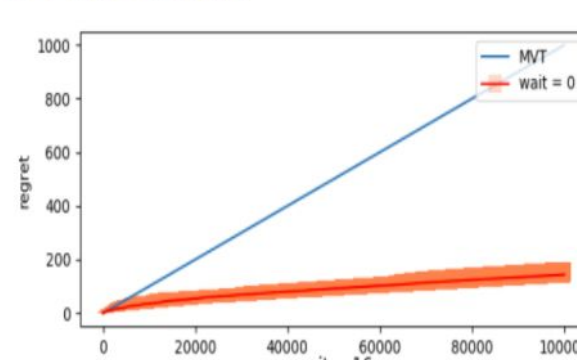
10% CR, 5% Uplift



10% CR, 10% Uplift



10% CR, 20% Uplift



In all cases, we use 10% traffic for exploration. We see in all cases, the cumulative regret gets saturated over time with the proposed algorithm.

Proposed solution



PROPOSED SOLUTION

Solution:

3 stages of enhancements

SOLUTION

Lot 1: Awareness

Goal: Complement current reporting with explainers and documentation to be more trustworthy and transparent

Proposed tier	First Effort evaluation	Timeline proposal
T1	2.5 Sprints	End of Q4

Solution Proposed

- Add a “Why this variant is preferred” panel to *Overview* or *Learnings* tab that summarizes rationale using algorithmic uncertainty, not statistical CI
- Fork approach for MAB and CB:
 - MAB: This is simply the variation performance, so simpler
 - Start by using a chart such as OPT for **Top User Attributes**
- Abundant tooltips for exploration, exploitation, and confidence terms used
- Publish or brush up 2–3 help-center articles to educate customers, using consistent terminology

Top User Attributes ?

#	Name	Impact	
1	Device type	27%	
2	Product type	8%	
3	User role	8%	
4	Language	6%	

SOLUTION

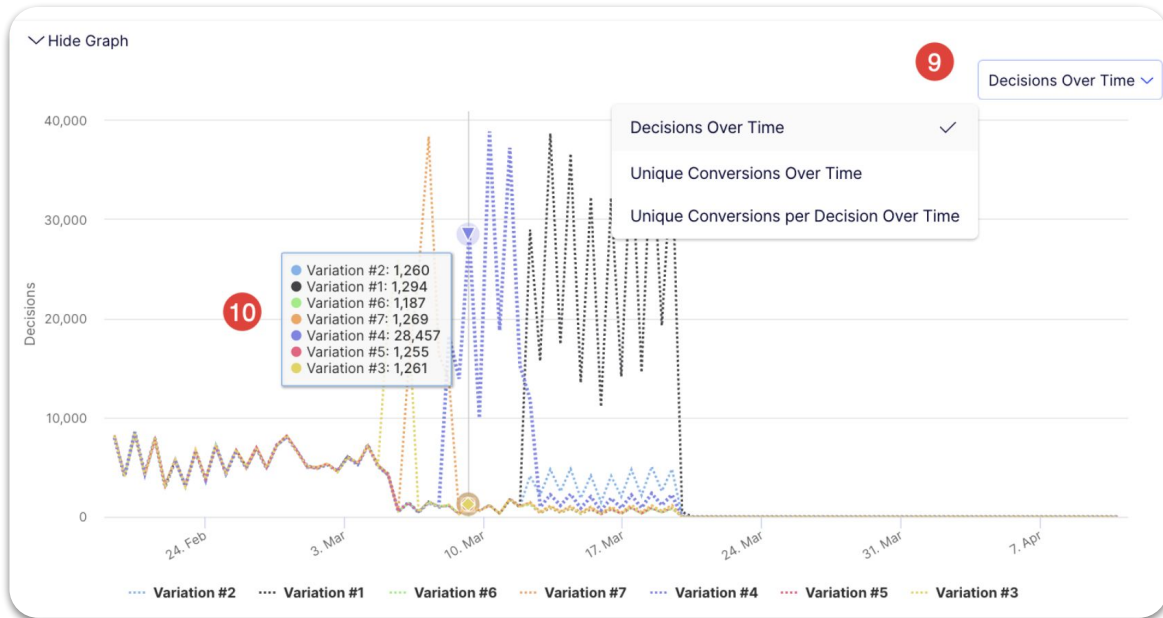
Lot 2: Reporting

Goal: Add time/variation graphs and replace invalid stats with new columns

Proposed tier	First Effort evaluation	Timeline proposal
T1	? Sprints	Q1 2026

Solution Proposed

- Initiate a light clean-up of the RP
- Remove Spider chart and introduce a traffic allocation timeline with model-update markers and hover details (like SS, OPT)
- Remove all mentions of reliability and confidence intervals (like OPT)



SOLUTION

Lot 3:
Governance

Goal: Measure the effectiveness of contextual bandits by running them side-by-side with a control (holdout XP)

Proposed tier	First Effort evaluation	Timeline proposal
T3	-	H1 2026

Solution Proposed

- Introduce a **Improvement vs Equal Allocation** approach for bandits
- Redo entirely how our backend engine measures, compares and stores CB data
- Initiate a **larger scale effort** to modularize the RP so that only relevant info displays

←

CONTEXTUAL BANDIT REPORTING 2.0

Experiment ID: 189380

Current Leading Variant Redesign 1 ①1	Estimated Improvement vs Equal Spilt ① +9,8 % (± 2,1%)	Cumulative Reward (All Variants) 5,016.646 €	Explorative Rate 12 %	Model Health ✔ Stable Last update
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Newsletter registration PRIMARY GOAL

Variation	Visitors	Conversion rate	Revenue	Revenue per visitor	Cumulative reward
● Original	431,307	1,88 %	1.025,108 €	3.017,257 €	4.042.365 €
● Redesign 1	559,646	2.57%	3.017,257 €	3.017.257 €	1.177,348 €
● Redesign 2	589,475	0,91 %	692.240 €	1.176 €	1.217.349 €
All variations	1,574,328	2,21%	5.016.646 €	3,18 €	

Charts

Revicletter registration

Traffic Allocation Over Time

Cumulative Reward by Segment

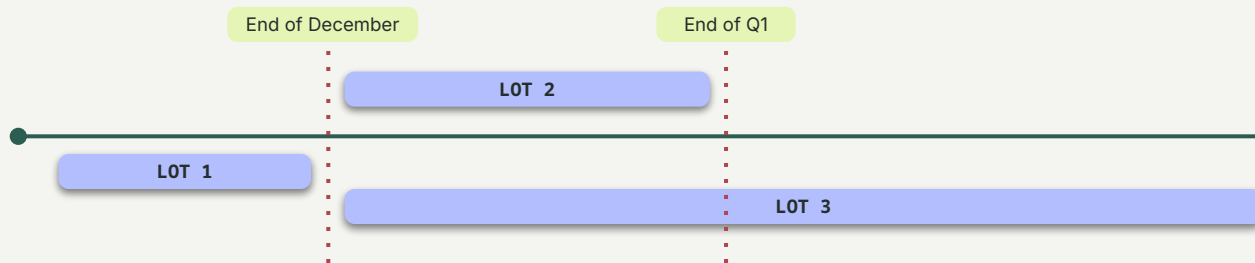
Mobile

Desktop

Tablet

How the Bandit Learned (View Learning Timeline)

Proposed Timeline



Open discussion

Decision

Thank You